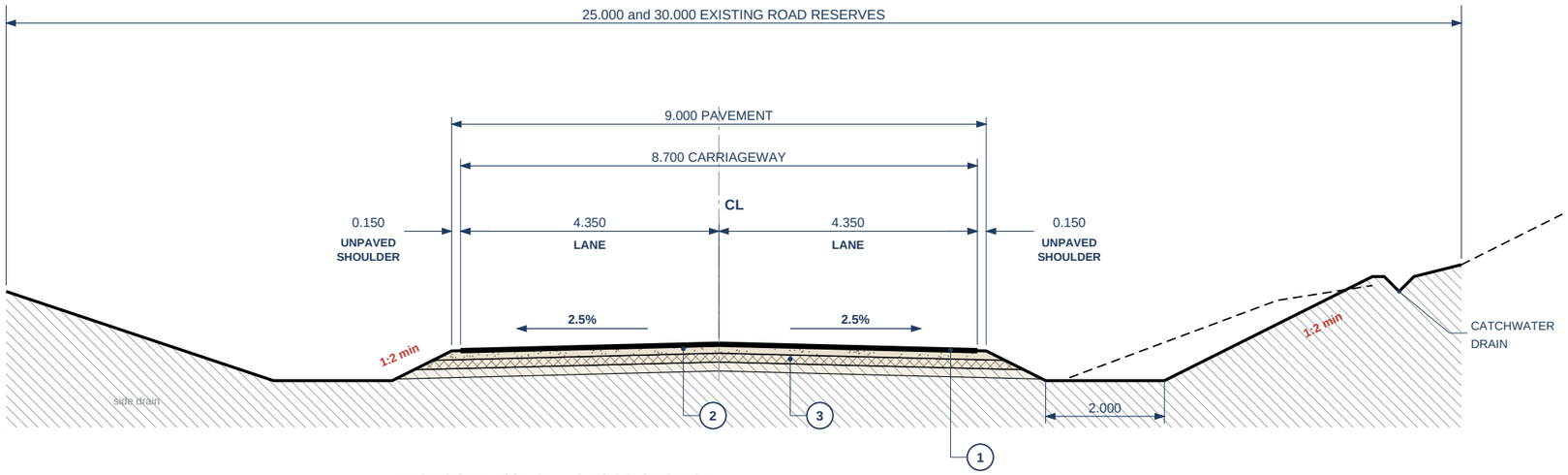


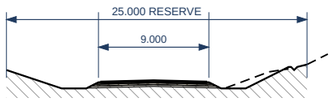
PROPOSED TYPICAL CROSS SECTION

Drawn at 1:100 for legibility (required 1:500 shown true-size below) · dimensions in metres

TABLE A : PAVEMENT STRUCTURE SPECIFICATIONS

LAYER	THICKNESS (mm)	DESCRIPTION	MATERIAL CODE	SATCC SPECIFICATIONS SECTION	COMPACTION % OF MOD AASHTO	PI MAX	GM MIN TO MAX	CBR% @ MOD AASHTO DENSITY	UCS (MPa)	ITS (MPa)	MATERIAL SOURCE
1	-	**Single Seal 13.2 mm (First Stage) 25mm asphalt (Second Stage) Binder 80/100 Bitumen grade**	S1	4400		-	-	-	-	-	Commercial
2	150	**Base**	G4	3400 + 3500	98	6 *	-	-	-	-	In-situ / Commercial
3	150	**Subbase**	G5	3400 + 3500	95	10 *	1.5 - 2.5	-	-	-	In-situ / Commercial
4	-	**Roadbed**	INSITU MATERIAL: SCARIFY 150mm AND COMPACT TO 93% MOD AASHTO DENSITY								In-situ /

IN EXPANSIVE CLAYS, CUT TO WASTE AND FILL



SECTION AT REQUIRED SCALE 1:500 (true size)

Use this size on the submitted sheet; read all values from the 1:100 view

0 5 m @ 1:500

0 5 m @ 1:100

NOTES

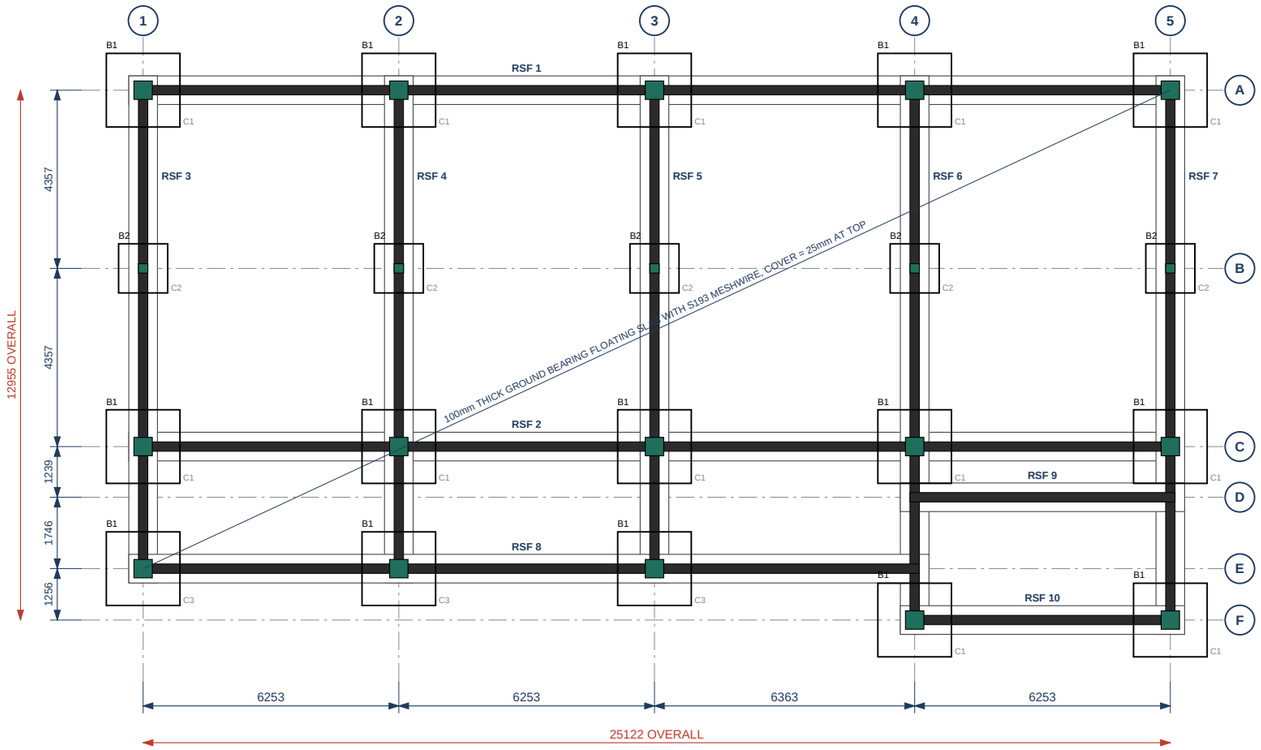
1. Carriageway 8.700 = 2 lanes x 4.350; unpaved shoulders 0.150 each side; pavement 8.700 + 2 x 0.150 = 9.000.
2. Crossfall 2.5% from CL; side slopes 1:2 min; side drain base 2.000; catchwater drain on cut side.
3. Layers 1-4 as Table A (callouts on section).
4. Dashed line = natural ground before cut.

ROAD CROSS-SECTION

Shouldered two-way carriageway · Q1

Institute	**NUST — Faculty of Engineering**	Course	**ECW1102**
Project title	**INDIVIDUAL ASSIGNMENT 3**	Scale	**1:500 (1:100 view)**
Drawn by	**Karl (student no.)**	Sheet	**A3**
Date	**05/10/2026**	Units	**mm / m (levels)**
Checked	**—**	Dwg no.	**IA3-Q1 / R-01**

Reproduced from Individual Assignment 3, Q1 (source: MOTID).
Widths in metres; layer thicknesses in mm.



FOUNDATION AND GROUND FLOOR SLAB LAYOUT

Scale 1:150 · grid dimensions as given; overall sizes (red) added = sum of chains

MEMBER	No. OFF	SIZE
BASE B2	5	1200 x 1200 x 350
COLUMN C2	5	230 x 230
BASE B1	20	1800 x 1800 x 450
COLUMN C1	12	450 x 450
COLUMN C3	3	450 x 450

BASE & COLUMN B/SCHEDULE

MEMBER & No.	BAR MARK	TYPE & SIZE	No. OF MBRS	No. IN EACH	TOTAL No.	LENGTH (mm)	SHAPE
B1; 5No (1200x1200x350)	01	Y16	5	14	70	1700	300 / 1100 / 300
C1; 5No (230x230)	02	Y16	5	4	20	2600	500 / 2100
	03	Y10	5	23	115	725	130 / 130
	07	Y16	5	4	20	3025	STR
B1; 20No (1800x1800x450)	01	Y16	20	18	360	2100	200 / 1700
COLUMN3; 3nos 450x450	05	Y16	3	8	24	2350	350 / 2000
	06	Y10	3	23	184	1600	350 / 350
	07	Y16	3	8	24	3025	STR
COLUMN: 1nos 450x450	05	Y16	12	8	24	2350	350 / 2000
	06	Y10	12	38	456	1600	350 / 350
	07	Y16	12	8	24	3025	STR
	07	Y16	12	8	24	3025	STR

B/SCHEDULE (REINFORCED FOOTING)

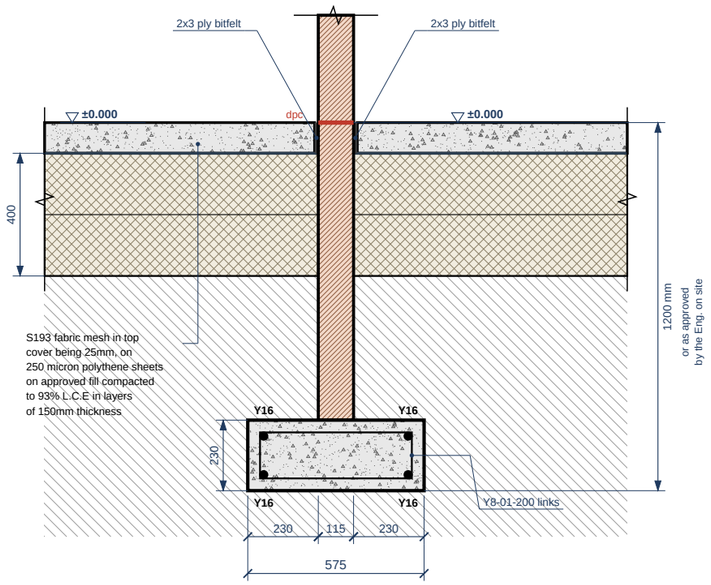
MEMBER & No.	BAR MARK	TYPE & SIZE	No. OF MBRS	No. IN EACH	TOTAL No.	LENGTH (mm)	SHAPE
RSF 1 (700x300) RSF 2	01	Y10	2	130	260	2000	250 / 650 Links
	02	Y16	2	18	36	9407	STR
RSF 3 (700x300) RSF 4,5	01	Y10	3	63	189	2000	250 / 650 Links
	02	Y16	3	12	39	7000	STR
RSF 6 (700x300) RSF 7	01	Y10	1	69	69	2000	250 / 650 Links
	02	Y16	1	12	12	7000	STR
RSF 8 (700x300)	01	Y10	1	100	100	2000	250 / 650 Links
	02	Y16	1	12	12	10600	STR
RSF 9 (700x300) RSF 10	01	Y10	2	13	13	2000	250 / 650 Links
	02	Y16	2	6	12	7000	STR

FOUNDATION LAYOUT & SCHEDULES

Double-storey structure · Q2 sheet 1 of 3

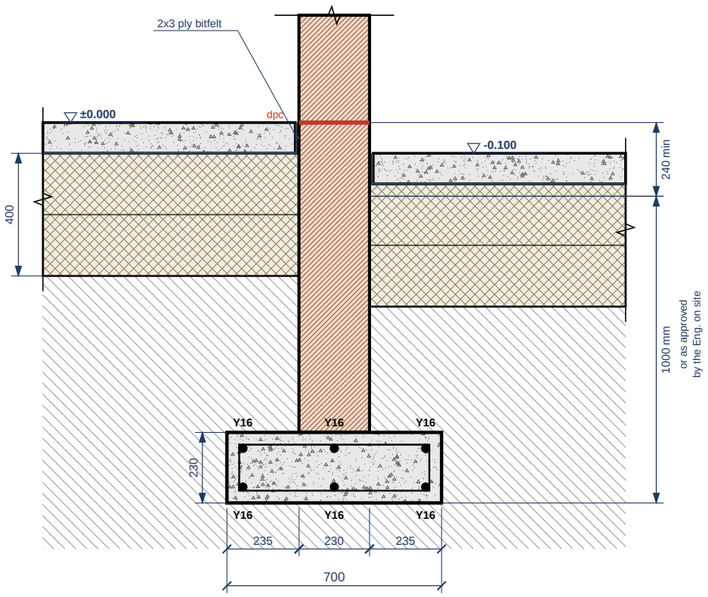
Institute	**NUST — Faculty of Engineering**	Course	**ECW1102**
Project title	**INDIVIDUAL ASSIGNMENT 3**	Scale	**1:150**
Drawn by	**Karl (student no.)**	Sheet	**A3**
Date	**05/10/2026**	Units	**mm / m (levels)**
Checked	**—**	Dwg no.	**IA3-Q2 / S-01**

Reproduced from Individual Assignment 3, Q2 (Raytech Consulting).
Dimensions in mm unless noted; levels in metres.



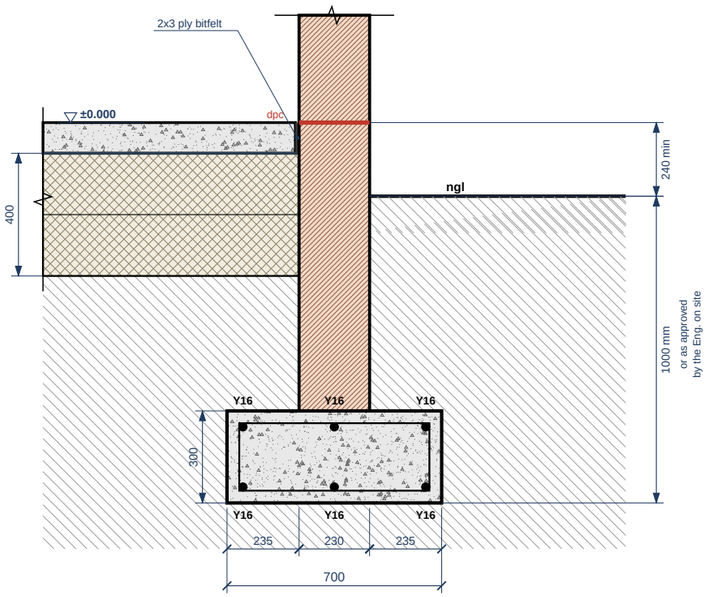
SECTION B-B

Scale 1:20 · internal 115 wall



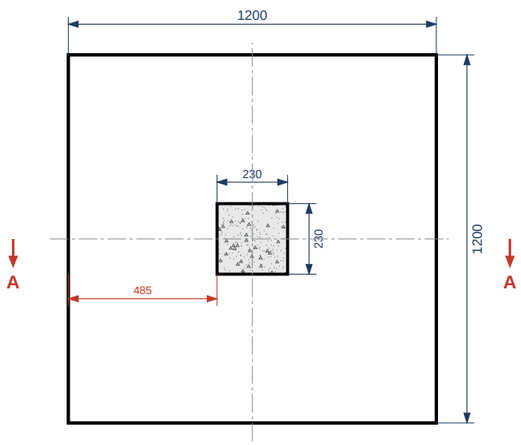
SECTION C-C

Scale 1:20 · 230 wall, floor step -0.100



SECTION A-A

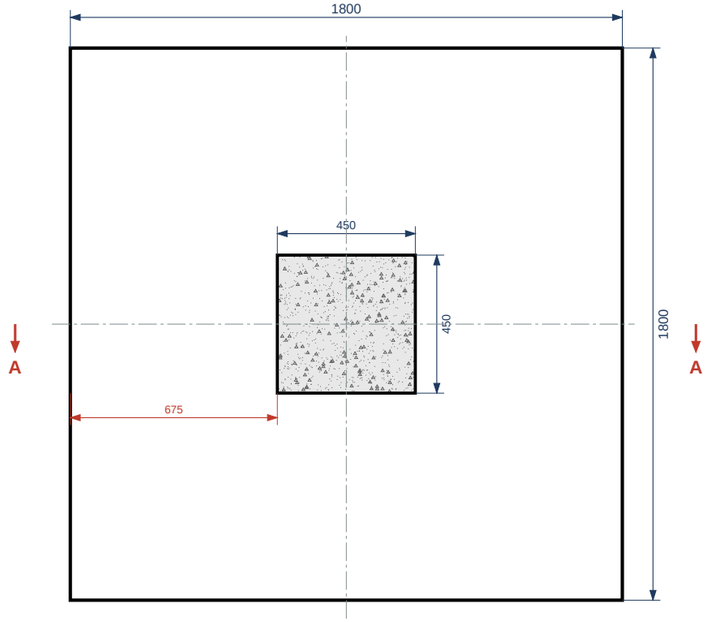
Scale 1:20 · external 230 wall



BASE B2 & COL C2 LAYOUT

BASE B2 = 1200x1200x350 5 Nos
COLUMN C2 = 230x230 5 Nos

Scale 1:20 · red = edge distance (added: (base - column) / 2)



BASE B1 & COL C1 LAYOUT

BASE B1 = 1800x1800x450 (20 Nos)
COLUMN C1 = 450x450 (12 Nos)

NOTES

1. All dimensions in mm; levels in metres.
2. 1200 / 1000 founding depths 'or as approved by the Engineer on site'.
3. 240 min = ±0.000 floor level to natural ground (ngl).
4. Footings: 575x230 (internal 115 walls), 700x230 and 700x300 (230 walls); main bars Y16, links Y8-01-200.
5. dpc at floor level; 2x3 ply bitfelt between slab and wall.
6. Dimensions in red are derived from the given sizes.
7. Slab (A-A, B-B, C-C): 100 thick, S193 mesh in top (25 cover), 250 micron polythene on fill compacted to 93% L.C.E in 150 layers.

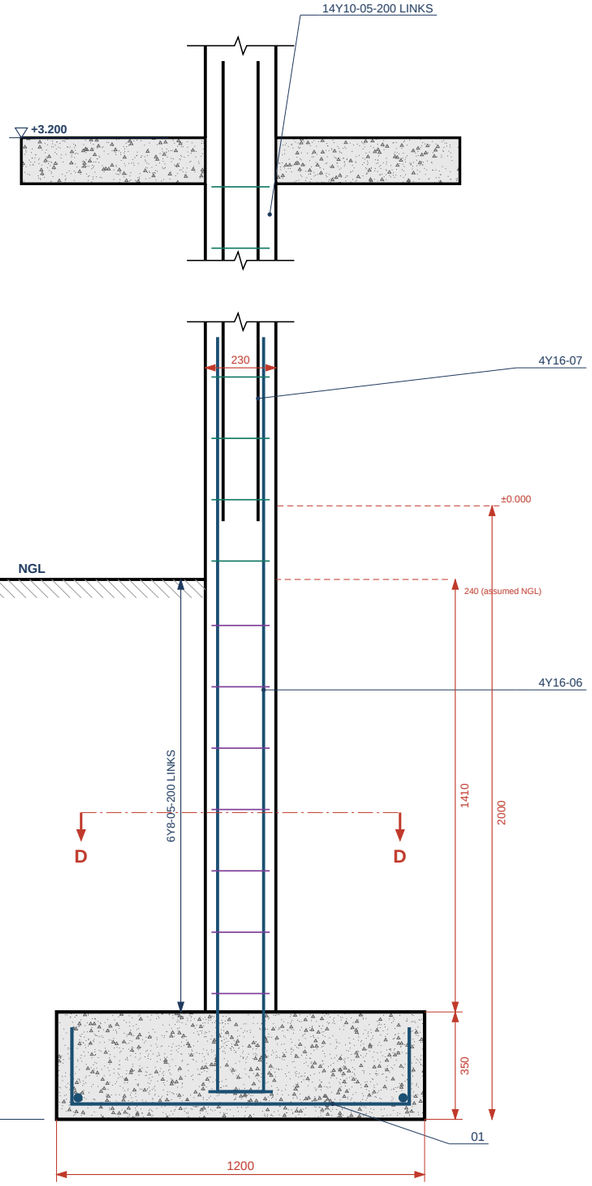
concrete · brickwork · compacted fill · natural ground

FOOTING SECTIONS & BASE LAYOUTS

Double-storey structure · Q2 sheet 2 of 3

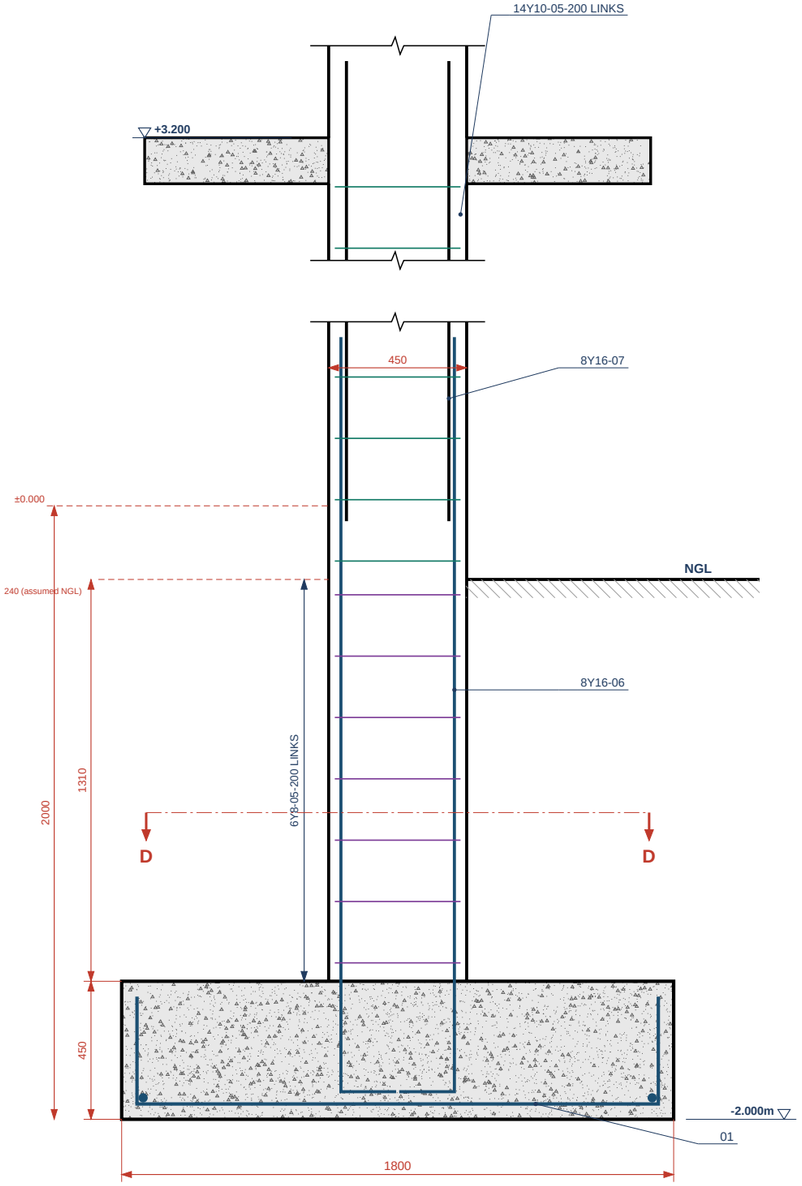
Institute	**NUST — Faculty of Engineering**	Course	**ECW1102**
Project title	**INDIVIDUAL ASSIGNMENT 3**	Scale	**1:20**
Drawn by	**Karl (student no.)**	Sheet	**A3**
Date	**05/10/2026**	Units	**mm / m (levels)**
Checked	**—**	Dwg no.	**IA3-Q2 / S-02**

Reproduced from Individual Assignment 3, Q2 (Raytech Consulting).
Dimensions in mm unless noted; levels in metres.



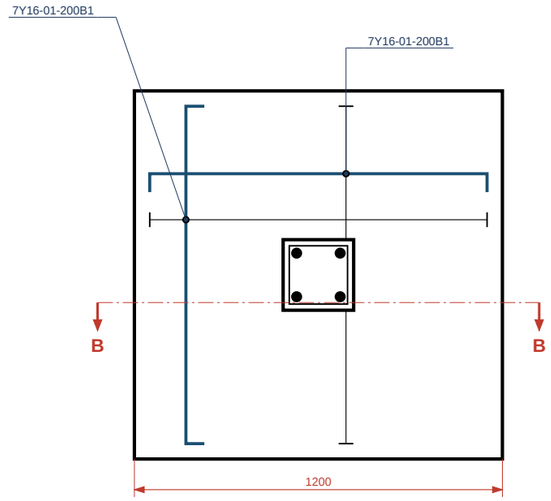
SECTION C-C — COLUMN C2 230 x 230 mm

Scale 1:20 (column shortened at break)



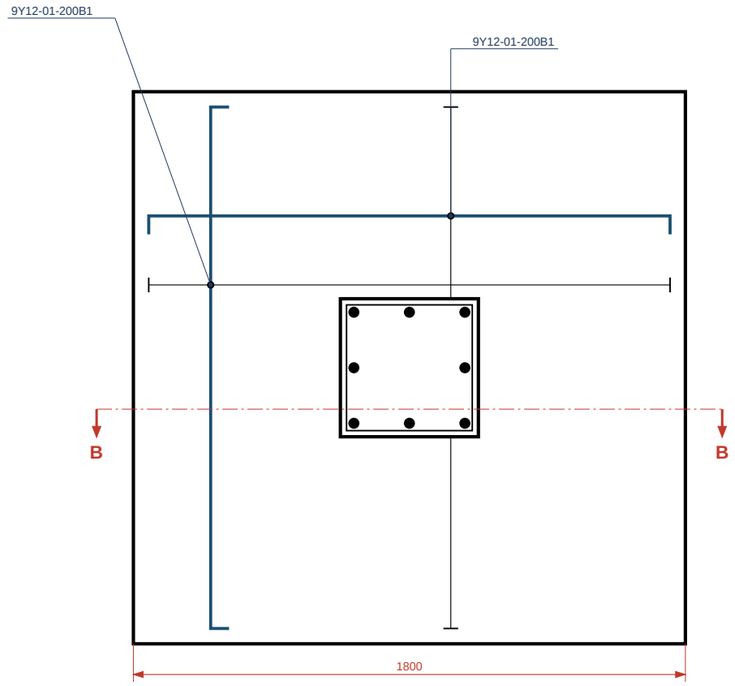
SECTION C-C — COLUMN 450 x 450 mm

Scale 1:20 (column shortened at break)



BASE B2 & COL C2 R.C DETAILS

Scale 1:20



BASE B1 & COL C1 R.C DETAILS

Scale 1:20

COLUMN SECTIONS & BASE R.C.

Double-storey structure · Q2 sheet 3 of 3

Institute	**NUST — Faculty of Engineering**	Course	**ECW1102**
Project title	**INDIVIDUAL ASSIGNMENT 3**	Scale	**1:20**
Drawn by	**Karl (student no.)**	Sheet	**A3**
Date	**05/10/2026**	Units	**mm / m (levels)**
Checked	**—**	Dwg no.	**IA3-Q2 / S-03**

Reproduced from Individual Assignment 3, Q2 (Raytech Consulting).
Dimensions in mm unless noted; levels in metres.